Congenital Heart Defects and Ectodermal Dysplasia (CHDED / TEBC Syndrome): A Comprehensive Disease Characterization

Disease: Congenital Heart Defects and Ectodermal Dysplasia (CHDED) **Causal gene:** *PRKD1* (protein kinase D1; HGNC:9407; NCBI Gene 5587; UniProt Q15139) **MONDO ID:** MONDO:0044303 · **OMIM:** #617364 **Synonyms:** CHDED; CHDFIDD; congenital heart defects, dysmorphic facial features, and intellectual developmental disorder; telangiectasia–ectodermal dysplasia–brachydactyly–cardiac anomaly (TEBC) syndrome; PRKD1-related syndrome; PRKD1 gain-of-function syndrome **Category:** Mendelian, autosomal dominant

Summary

Congenital Heart Defects and Ectodermal Dysplasia (**CHDED**, also called **TEBC syndrome**) is an ultra-rare, autosomal dominant, multisystem developmental disorder caused by heterozygous — usually *de novo* — pathogenic variants in **PRKD1**, the gene encoding **protein kinase D1**, a diacylglycerol/PKC-activated serine–threonine kinase. The disorder is defined by the co-occurrence of **congenital heart defects** and **ectodermal dysplasia**, frequently accompanied by **telangiectasia**, **brachydactyly/skeletal anomalies**, **facial dysmorphism**, and variable **neurodevelopmental, renal, endocrine, and cerebrovascular** features. Only approximately 8–10 molecularly confirmed patients had been reported in the literature as of 2024–2025, so all disease-level knowledge derives from aggregated individual case reports and functional/model-organism studies rather than population registries.

A central, well-supported conclusion of this investigation is that *PRKD1* exhibits a **dual genetic architecture**. **Biallelic truncating (loss-of-function) variants** cause an isolated recessive outflow-tract malformation (**truncus arteriosus**), first mapped in consanguineous families, whereas **heterozygous missense (and in-frame splice) variants clustered in the catalytic kinase domain** cause the dominant syndromic disorder CHDED. Population constraint data (gnomAD pLI ≈ 0; LOEUF ≈ 0.81) indicate that simple haploinsufficiency is largely tolerated,

arguing that the dominant syndrome operates through a **gain-of-function / dominant-negative** mechanism — a conclusion reinforced by newer clinical-genetics reports that explicitly term the disorder a "**PRKD1 gain-of-function syndrome**."

Mechanistically, PRKD1 functions downstream of PKC/diacylglycerol and directly phosphorylates the class IIa histone deacetylase **HDAC5**, driving its nuclear export and thereby **derepressing MEF2** transcription factors that govern cardiac gene programs. A parallel, proposed branch links PRKD1 to an **endothelial ITGB2/JAM2 tight-junction pathway** implicated in brain and organ calcification. GTEx expression data show PRKD1 is enriched in **arteries, fibroblasts, nerve, and renal medulla**, mirroring the vascular/outflow-tract, ectodermal, and renal phenotypes. Diagnosis is genotype-driven (trio exome/genome sequencing); there is **no disease-specific or molecularly targeted therapy**, and management is entirely supportive and multidisciplinary.

Key Findings

F001 — CHDED is an autosomal dominant syndrome caused by *PRKD1* variants

CHDED (MIM #617364; MONDO:0044303) is an **autosomal dominant** disorder caused by **heterozygous pathogenic variants in PRKD1**, characterized mainly by congenital heart defects and ectodermal dysplasia with variable skeletal, neurodevelopmental, facial, and other features. It is also known as **TEBC** (telangiectasia–ectodermal dysplasia–brachydactyly–cardiac anomaly) syndrome and **CHDFIDD**. Most dominant cases arise from *de novo* missense variants, and only ~8–10 patients were described as of 2024.

> *"Congenital heart disease and ectodermal dysplasia syndrome (CHDED syndrome) (MIM: 617364) is an autosomal dominant disorder cause by PRKD1 gene pathogenic variants, characterised mainly by congenital heart defects (CHD) and ectodermal dysplasia, along with other variable clinical features (including skeletal defects)."* — PMID: 41136760
>
> *"Telangiectasia-ectodermal dysplasia-brachydactyly-cardiac anomaly (TEBC) syndrome is a rare autosomal dominant condition, recently linked to the protein kinase D1 (PRKD1) gene."* — PMID: 38677542

F002 — Dual genetic architecture: dominant missense (syndromic) vs. recessive truncating (isolated truncus arteriosus)

Shaheen et al. (2015) first mapped *PRKD1* by autozygosity in two consanguineous Saudi families with **truncus arteriosus (TA)**, identifying **homozygous truncating mutations** — a recessive loss-of-function mechanism causing isolated outflow-tract defects. In contrast, syndromic CHDED/TEBC is caused by **heterozygous (usually *de novo*) missense variants** acting via a gain-of-function/dominant mechanism (e.g., p.Arg603His / c.1808G>A; p.Val712Met / c.2134G>A). Newer reports explicitly describe a "PRKD1 Gain-of-Function Syndrome." A novel splice variant (c.1906-1G>T) causing exon 14 skipping was reported in a familial case.

> *"we demonstrate in two multiplex consanguineous families that we are able to map TA to regions of autozygosity in which whole-exome sequencing revealed homozygous truncating mutations in PRKD1 (encoding a kinase derepressor of MAF2) and NRP1"* — PMID: 25713110
>
> *"The patient had a de novo, heterozygous pathogenic variant, c.1808G > A, p.(Arg603His), in the PRKD1 gene."* — PMID: 39740729

Feature	Recessive form	Dominant form (CHDED/TEBC)
Zygosity	Biallelic (homozygous)	Heterozygous
Variant type	Truncating (LoF)	Missense / in-frame splice
Mechanism	Loss of function	Gain-of-function / dominant-negative
Phenotype	Isolated truncus arteriosus	Multisystem syndrome
Typical origin	Consanguinity	*De novo* (rarely inherited)
Landmark ref.	PMID: 25713110	PMID: 27479907

F005 — *PRKD1* established as a genome-wide-significant syndromic CHD gene (Sifrim 2016)

Sifrim et al. (2016, *Nature Genetics*) exome-sequenced **1,891 CHD probands** (610 syndromic, 1,281 nonsyndromic) and identified **three genome-wide-significant syndromic-CHD disorders** caused by *de novo* mutations in **CHD4, CDK13, and PRKD1**. Syndromic CHD showed enrichment of *de novo* protein-truncating variants in known CHD genes, whereas

nonsyndromic CHD showed enrichment of inherited PTVs — evidence for "distinct genetic architectures." This is the landmark study establishing *PRKD1* as a cause of dominant syndromic CHD, complementing the earlier recessive TA mapping.

> *"We identified three genome-wide significant S-CHD disorders caused by DNMs in CHD4, CDK13 and PRKD1."* — PMID: 27479907
>
> *"Our study finds evidence for distinct genetic architectures underlying the low sibling recurrence risk in S-CHD and NS-CHD."* — PMID: 27479907

F004 — Phenotype spectrum: four cardinal features plus neurodevelopmental, renal, and cerebrovascular involvement

Across the ~8–10 reported patients, the **core tetrad** comprises:

1. **Congenital heart defects** — atrioventricular septal defect (partial/complete; HP:0006695), pulmonary/pulmonic stenosis (HP:0001642), coarctation of the aorta (HP:0001680), bicuspid aortic valve (HP:0001647), truncus arteriosus, tetralogy of Fallot (HP:0001636), partial anomalous pulmonary venous return (PAPVR);
2. **Ectodermal dysplasia** (HP:0000968) affecting hair, nails, teeth, and skin;
3. **Telangiectasia** of childhood onset (HP:0007394);
4. **Brachydactyly / skeletal defects** (HP:0001156).

Additional variable features include facial dysmorphism (HP:0001999), intellectual disability (HP:0001249) with microcephaly (HP:0000252), multicystic renal dysplasia (HP:0000003), nephrocalcinosis/renal calcification (HP:0000121), cerebral (deep white-matter) calcification (HP:0002514) with seizures (HP:0001250), increased intracranial pressure, bifid uvula (HP:0000193), sagittal craniosynostosis (HP:0004440), hormonal resistance, and (in an adult) bilateral carotid artery dissection.

> *"The proband presents with partial atrioventricular septal defect, brachydactyly, ectodermal dysplasia, telangiectasia that developed in childhood, intellectual disability with microcephaly, multicystic renal dysplasia and moderate hormonal resistance."* — PMID: 38677542

> *"Individuals with a heterozygous mutation in PRKD1 may have facial dysmorphism, ectodermal dysplasia and may have CHDs such as pulmonary stenosis, atrioventricular septal defects, coarctation of the aorta and bicuspid aortic valve."* — PMID: 38419169
>
> *"Previously unreported features were noted including high intracranial pressure (ICP), partial anomalous pulmonary venous return (PAPVR), and bifid uvula."* — PMID: 36308391

F009 — Pathogenic missense variants cluster in the catalytic kinase domain

PRKD1 (UniProt Q15139, 912 aa) comprises two phorbol-ester/DAG-binding **C1 zinc fingers** (~residues 146–196 and 270–320), a **pleckstrin-homology (PH) domain** (~422–541), and a C-terminal **serine/threonine kinase domain** (~583–839), with the catalytic proton-acceptor active site at Asp706 and activation-loop autophosphorylation/PKC sites at Ser738/Ser742 (also reported as Ser744/Ser748). Reported pathogenic CHDED missense variants map **inside the kinase domain**: **p.Arg603His** (c.1808G>A), **p.Val712Met** (c.2134G>A), and the splice variant **c.1906-1G>T**, which causes skipping of exon 14 within the kinase-domain-encoding region. Relevant GO annotations: GO:0004672 (protein kinase activity), GO:0004697 (diacylglycerol-dependent serine/threonine kinase activity), GO:0005524 (ATP binding), GO:0006468 (protein phosphorylation).

> *"We identified a novel heterozygous splice variant in PRKD1 (c.1906-1 G > T) in the proband, inherited from the affected father. Transcript analysis confirmed that the PRKD1 splice variant caused the complete skipping of exon 14."* — PMID: 41136760
>
> *"We report on the 8th patient carrying a heterozygous de novo variation of PRKD1 c.2134G > A, p. (Val712Met)"* — PMID: 38677542

F008 — gnomAD constraint supports gain-of-function, not haploinsufficiency

PRKD1 (ENSG00000184304; HGNC:9407; chr14q12; GRCh38 14:29,576,479–30,191,898, minus strand; canonical transcript ENST00000331968 with 18 exons; protein 912 aa). gnomAD v2 constraint metrics: **pLI ≈ 0** (8.5e-14), **observed/expected LoF = 0.66** (LOEUF oe_lof_upper = 0.81; obs 63 vs. exp 95.9 LoF variants), **missense Z = 1.47** (oe_mis = 0.90), **LoF Z = 2.85**. PRKD1 is only moderately LoF-constrained and is **not predicted to be haploinsufficient** (pLI ≈ 0): simple heterozygous loss of function is largely tolerated in the general population, consistent with a gain-of-function/dominant-negative basis for the dominant syndrome.

> *"Expanding the Phenotypic Spectrum of PRKD1 Gain-of-Function Syndrome"* — PMID: 42792942 (title framing independently supports the gain-of-function inference)

F003 — Core molecular mechanism: PKC → PKD1 → HDAC5 phosphorylation/ nuclear export → MEF2 derepression

PRKD1 encodes a serine/threonine protein kinase activated downstream of PKC/DAG via activation-loop phosphorylation (Ser744/Ser748). A core validated function is **direct phosphorylation of class IIa histone deacetylase HDAC5**, driving its CRM1-dependent **nuclear export** and thereby **derepressing MEF2** transcription factors that control cardiac gene programs. Shaheen et al. describe PRKD1 as "a kinase derepressor of MEF2." *Prkd1*-null mice are embryonic lethal, and tissue-specific deletion causes abnormal cardiac remodeling. A gain-of-function missense variant would be expected to **increase** HDAC5 phosphorylation and MEF2 activity.

> *"protein kinase D (PKD), a downstream effector of PKC, directly phosphorylates HDAC5 and stimulates its nuclear export"* — PMID: 15367659
>
> *"Previous work has demonstrated that Prkd1(-/-) is embryonic lethal and that its tissue-specific deletion results in abnormal heart remodelling"* — PMID: 25713110

F006 — Brain/organ calcification via a proposed PRKD1–ITGB2–JAM2 endothelial tight-junction pathway

Watanabe et al. (2025) reported a 9-month-old girl with CHDED (*de novo* p.Arg603His) presenting with heart defects, ectodermal dysplasia, seizures, and **bilateral deep cerebral white-matter calcifications** (seizures controlled with levetiracetam). Reviewing prior cases, calcifications occurred in **3 of 8 patients** (2 renal, 1 brain). The authors propose that PRKD1 acts in a **common pathway with ITGB2 and JAM2** — both established primary familial brain calcification genes — where PRKD1 partners with integrin ITGB2 and JAM2 maintains endothelial tight junctions, implicating **neurovascular unit / blood–brain-barrier dysfunction**.

> *"We propose that PRKD1 and two genes, ITGB2 and JAM2, which are known to be associated with brain calcification, act through a common signaling pathway abnormality."* — PMID: 39740729
>
> *"At 6 months of age, she had generalized seizures, and a CT scan revealed calcifications in the bilateral deep cerebral white matter."* — PMID: 39740729

F012 — PRKD1 expression is enriched in arterial/vascular, fibroblast, nerve, and renal-medulla tissue

GTEx v8 median expression shows *PRKD1* is highest in **cultured fibroblasts (45.8 TPM)** and **arteries** — aorta 24.8, tibial 21.3, coronary 19.4 TPM — plus testis (22.2), tibial nerve (21.6), prostate (21.5), **kidney medulla (20.1)**, ovary (19.3), and bladder (18.7), but **low in adult heart left ventricle (1.4 TPM)**, skeletal muscle (0.9), liver (0.8), and whole blood (0.04). The strong arterial/vascular-smooth-muscle, fibroblast (mesenchymal), and renal-medulla expression parallels the disease's outflow-tract/arterial heart defects (truncus arteriosus, coarctation, bicuspid aortic valve), cutaneous telangiectasia, carotid artery dissection, and renal dysplasia/nephrocalcinosis.

> *"Congenital Heart Disease, Bilateral Carotid Dissections, and Ectodermal Dysplasia in an Adult Patient"* — PMID: 42792942 (vascular manifestation matches PRKD1's high arterial expression)

Affected anatomy (UBERON): heart (UBERON:0000948), cardiac outflow tract (UBERON:0004145), aorta (UBERON:0000947), skin (UBERON:0002097), kidney (UBERON:0002113), brain (UBERON:0000955), teeth (UBERON:0001091), hair follicle (UBERON:0002073). **Cell types (CL):** endothelial cell (CL:0000115), vascular smooth muscle cell (CL:0000359), fibroblast (CL:0000057), cardiac neural crest-derived cells, keratinocyte (CL:0000312).

F007 — *Prkd1* mouse models recapitulate cardiac developmental defects

Prkd1 knockout is embryonic lethal, and tissue-specific deletion produces abnormal heart remodeling. Waheed-Ullah et al. (2024) analyzed a *Prkd1* transgenic mouse model to study its role in cardiogenesis and CHD, framing PRKD1 as "an essential protein kinase" important in cardiac morphogenesis. Mouse *Prkd1* (NCBI Gene 18760) is the ortholog of human *PRKD1* (NCBI Gene 5587; HGNC:9407); the kinase is broadly conserved.

> *"To obtain a greater appreciation for the role that this essential protein kinase plays in cardiogenesis and CHD, we have analysed a Prkd1 transgenic mouse model"* — PMID: 38419169

F011 — Epidemiology and inheritance: ultra-rare, mostly *de novo*, with rare familial transmission and variable expressivity

CHDED/TEBC is **ultra-rare**: only ~8–10 molecularly confirmed patients reported as of 2024–2025; no population prevalence/incidence estimates exist. Inheritance is **autosomal dominant**. Most dominant cases are *de novo* heterozygous missense (e.g., p.Arg603His, p.Val712Met), but **vertical transmission occurs** — a familial case showed an affected father transmitting a splice variant (c.1906-1G>T) to an affected son with differential manifestation (TOF + sagittal craniosynostosis in the son vs. septal defects only in the father), demonstrating **intrafamilial variable expressivity**. No sex predilection, founder effect, anticipation, or carrier-frequency data are established; the separate recessive TA form was enriched by consanguinity in Saudi families.

> *"To date, eight patients with CHDED have been described."* — PMID: 39740729
>
> *"born of non-consanguineous Asian Indian origin couple with affected father and unaffected mother"* — PMID: 41136760

F010 — Management is symptomatic/multidisciplinary; no targeted therapy exists

No CHDED/TEBC-specific or PRKD1-targeted therapy, clinical trial, gene therapy, or approved drug exists (ultra-rare; no NCT trials identified). Management is **supportive and organ-directed**: surgical/interventional repair of congenital heart defects (AVSD/VSD closure,

coarctation repair, valve management); antiepileptic drugs for seizures (**levetiracetam** was effective and resolved seizures in the brain-calcification patient); dental, dermatologic, and hair/nail care for ectodermal dysplasia; neurodevelopmental/early-intervention and physical/occupational/speech therapy for developmental delay; nephrology surveillance for renal dysplasia/calcification; neurosurgical management of raised intracranial pressure/craniosynostosis; and vascular surveillance given reported carotid dissection in an adult. Genetic counseling is indicated.

> *"The seizures resolved with the administration of levetiracetam."* — PMID: 39740729

F014 — Diagnosis rests on trio exome/genome sequencing plus multi-organ imaging; no biomarker exists

All reported CHDED diagnoses were **molecular**, via trio whole-exome sequencing (WES) identifying *de novo*/inherited heterozygous *PRKD1* variants, often after SNP-microarray excluded CNVs including the **22q11.2 deletion**. WES/WGS and multigene congenital-heart/syndromic-CHD panels (which include *PRKD1* alongside *CDK13*, *CHD4*) are the recommended approach; single-gene *PRKD1* testing applies when the phenotype is recognized. Supporting phenotyping includes echocardiography/cardiac imaging; brain/renal CT or ultrasound for calcifications; EEG for seizures; renal ultrasound for multicystic dysplasia; skeletal radiographs for brachydactyly/craniosynostosis; and dental/dermatologic exam. **No specific laboratory biomarker** exists; diagnosis is genotype-driven. Differential diagnoses include 22q11.2 deletion/DiGeorge, other ectodermal dysplasias, Adams–Oliver and TEBC-overlapping syndromes, and PHACE/other CHD syndromes.

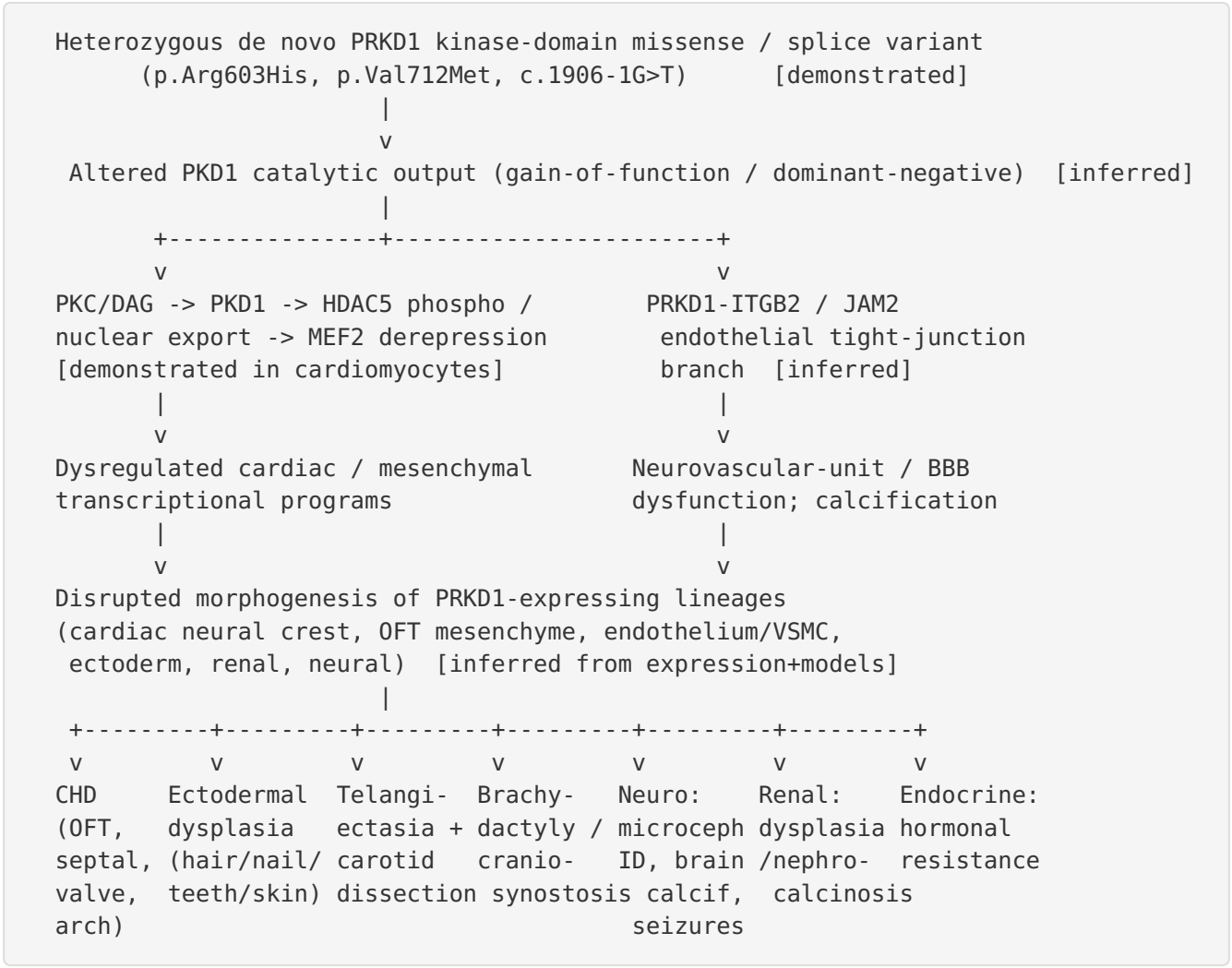
> *"Whole exome sequencing was performed on a 7-year-old male proband with CHD"* — PMID: 41136760
>
> *"molecular karyotyping to exclude 22q11.2 deletion syndrome"* — PMID: 25713110

Mechanistic Model / Interpretation

Integrated causal chain (F013)

The following ordered chain runs from the initiating genetic lesion to the clinical manifestations. Steps are marked **[demonstrated]** (directly shown experimentally) or **[inferred]** (plausible but not yet proven for patient variants).

1. A **heterozygous, usually *de novo* missense variant** in the PRKD1 kinase domain (e.g., p.Arg603His, p.Val712Met) or an **in-frame splice alteration** (exon-14 skipping) arises. **[demonstrated: human genetics — PMID: 27479907, 39740729, 41136760]**
2. This **alters PRKD1 serine/threonine kinase catalytic output** — gain-of-function/dominant-negative rather than haploinsufficiency (supported by gnomAD pLI ≈ 0 and "gain-of-function syndrome" framing). **[inferred]**
3. Dysregulated **PKC/DAG → PKD1 signaling** perturbs direct substrate phosphorylation, canonically **class IIa HDAC5 phosphorylation and CRM1-dependent nuclear export**. **[demonstrated in cardiomyocytes — PMID: 15367659]**
4. Altered HDAC5 export changes **MEF2-dependent transcriptional programs** ("PRKD1 a derepressor of MEF2"). **[demonstrated in model systems; inferred for patient variants]**
5. In parallel, PRKD1 dysfunction disturbs an **endothelial ITGB2/JAM2 tight-junction pathway**. **[inferred — PMID: 39740729]**
6. These transcriptional/cytoskeletal/junctional perturbations **disrupt morphogenesis of PRKD1-expressing lineages** — cardiac neural crest and second-heart-field/outflow-tract mesenchyme, vascular endothelium/smooth muscle, ectodermal appendages, and renal/neural tissue. **[inferred from expression + model data]**
7. **Branching to clinical manifestations** (see diagram below).



Upstream vs. downstream. The mutation and altered kinase activity are the most upstream events. The **PKD1→HDAC5→MEF2** axis and the **ITGB2/JAM2** branch are midstream signaling nodes. The tissue-specific morphogenetic failures and resulting malformations are the downstream clinical endpoints.

Key GO terms: GO:0007507 (heart development), GO:0003151 (outflow tract morphogenesis), GO:0001525 (angiogenesis), GO:0016477 (cell migration), GO:0006468 (protein phosphorylation), GO:0000122 (negative regulation of transcription). **Subcellular (GO CC):** GO:0005829 (cytosol), GO:0005794 (Golgi apparatus), GO:0005634 (nucleus), GO:0005886 (plasma membrane).

Final synthesis (F015)

Integrating all findings: **CHDED/TEBC** (MONDO:0044303, OMIM #617364, HGNC:9407 *PRKD1*, chr14q12) is autosomal dominant and ultra-rare (~8–10 patients). Pathogenic heterozygous missense/splice variants cluster in the **catalytic kinase domain** (aa ~583–839; p.Arg603His, p.Val712Met, c.1906-1G>T); gnomAD constraint (pLI ≈ 0, LOEUF ≈ 0.81) argues against haploinsufficiency and for **gain-of-function/dominant-negative**, contrasting with biallelic truncating LoF causing recessive isolated truncus arteriosus. Mechanism: **PKC/DAG → PKD1 → HDAC5 nuclear export → MEF2 transcription** (demonstrated in cardiomyocytes) plus a parallel **endothelial ITGB2/JAM2** branch; GTEx shows PRKD1 enriched in **arteries/fibroblasts/kidney-medulla** matching the outflow-tract, vascular, ectodermal, and renal phenotypes. Diagnosis is by trio WES/WGS; management is supportive; mouse *Prkd1* knockout is embryonic-lethal with cardiac remodeling defects.

Evidence Base

PMID	Title (abbrev.)	Role in this report
25713110	Positional mapping of PRKD1, NRP1, PRDM1 in truncus arteriosus	Establishes recessive LoF arm; PRKD1 as MEF2 derepressor; mouse KO lethality; 22q11.2 exclusion
27479907	Distinct genetic architectures for syndromic/nonsyndromic CHD (Sifrim 2016, *Nat Genet*)	Landmark: PRKD1 genome-wide-significant dominant syndromic CHD gene by *de novo* mutation
15367659	PKC/PKD mediate cardiac hypertrophy via HDAC5 nuclear export	Demonstrates the core PKD1→HDAC5→MEF2 mechanistic step
38677542	PRKD1-related TEBC syndrome: case report & review	Phenotype tetrad + neuro/renal/endocrine; 8th patient (p.Val712Met)
39740729	Brain calcification in CHDED	p.Arg603His; ITGB2/JAM2 branch; levetiracetam; case count (8 patients)
41136760	TOF & craniosynostosis in familial CHDED (novel splice variant)	Familial AD transmission; c.1906-1G>T exon-14 skipping; variable expressivity; WES
38419169	Deletion of PRKD1 in mouse embryonic heart	CHD types; *Prkd1* transgenic mouse model; "essential protein kinase"
36308391	Increased intracranial pressure in CHDED	Phenotype expansion: ICP, PAPVR, bifid uvula
42792942	Expanding PRKD1 Gain-of-Function Syndrome (adult)	"Gain-of-function" framing; carotid dissections; vascular manifestation

Supporting mechanistic literature (PKD1 biology). A substantial body of cardiomyocyte and lymphocyte signaling work corroborates the PKD1→HDAC5→MEF2 axis and PKD1 activation requirements: PKC-dependent activation-loop phosphorylation at Ser744/Ser748 (PMID: 15590638); spatiotemporal PKD activation and HDAC5 export in adult cardiomyocytes (PMID: 21795686, PMID: 22581927); AKAP-Lbc scaffolding of PKD→HDAC5 in hypertrophy (PMID: 18951085); and PKD/CaMKIIδ upregulation redistributing HDAC5 in heart failure (PMID: 18218981). These papers were used to understand normal PKD1 function; they concern hypertrophy/heart-failure signaling in adult myocardium rather than the developmental malformation that defines CHDED, so they support the mechanistic model indirectly.

Note on evidence source types. Disease-defining evidence is **human clinical** (case reports and the Sifrim exome cohort). Mechanistic evidence is **in vitro** / **model-organism** (cardiomyocyte signaling, mouse knockouts). Constraint and expression inferences are **computational** (gnomAD, GTEx). The gain-of-function conclusion is an **inference** triangulated from constraint data and the clinical "gain-of-function syndrome" framing, not yet a direct biochemical demonstration for patient variants.

Section-by-Section Coverage

1. Disease information. See F001. Identifiers: MONDO:0044303; OMIM #617364; HGNC:9407 (*PRKD1*). Orphanet/ICD-11 entries are not firmly recorded for this ultra-rare entity. Information is derived from **aggregated individual case reports**, not EHR/registry data.

2. Etiology. Primary cause is **genetic** — heterozygous *de novo* (rarely inherited) *PRKD1* missense/splice variants (F001, F002). No established environmental risk or protective factors; no gene–environment interactions documented. The only "modifier"-type observation is intrafamilial variable expressivity (F011).

3. Phenotypes. See F004, with HPO terms and the one quantified frequency available (calcification in 3/8 patients). Onset is **congenital** for heart/skeletal features; telangiectasia develops in childhood; seizures/ICP are childhood-onset; carotid dissection was adult-onset. Severity and quality-of-life impact are variable and dominated by cardiac and neurodevelopmental burden; no formal QoL instrument data exist for this disorder.

4. Genetic/molecular information. Causal gene *PRKD1* (F001, F008, F009). Variant types: missense (p.Arg603His, p.Val712Met), in-frame splice (c.1906-1G>T → exon-14 skipping) for the dominant syndrome; truncating (biallelic) for the recessive form. Functional consequence: likely gain-of-function/dominant-negative (dominant) vs. loss-of-function (recessive). Germline origin; no somatic/cancer association relevant here. No established modifier genes; no epigenetic or chromosomal-abnormality associations beyond excluding 22q11.2.

5. Environmental information. Not applicable — no toxic, lifestyle, or infectious contributors identified for this Mendelian disorder.

6. Mechanism/pathophysiology. Full causal chain in the Mechanistic Model section (F003, F006, F013).

7. Anatomical structures affected. Organ level: heart (outflow tract, septa, valves, aortic arch), skin/ectodermal appendages, arteries/vasculature, kidney, brain, skeleton, endocrine axes. UBERON/CL terms listed in F012.

8. Temporal development. Congenital onset of structural defects; childhood emergence of telangiectasia, seizures, calcifications; the critical window is **embryonic cardiac/neural-crest morphogenesis**. The structural malformation course is non-progressive once formed, but complications (ICP, seizures, vascular dissection) can be progressive/episodic.

9. Inheritance and population. See F011. Autosomal dominant, mostly *de novo*; penetrance appears high but with **variable expressivity**; no prevalence/incidence, founder, anticipation, or carrier-frequency data.

10. Diagnostics. See F014. Trio WES/WGS; syndromic-CHD gene panels; confirmatory cardiac/renal/brain imaging and EEG; no biomarker.

11. Outcome/prognosis. No formal survival data. Prognosis is driven by severity of CHD (and its surgical repair) and by neurodevelopmental/renal/vascular complications. Recovery of structural defects is surgical; neurodevelopmental deficits persist.

12. Treatment. See F010. Supportive and multidisciplinary; NCIT concepts: surgical repair of cardiac defect, anticonvulsant therapy (levetiracetam), supportive care, genetic counseling. No pharmacogenomic or targeted options.

13. Prevention. No primary prevention (genetic etiology). Secondary/tertiary prevention via early cardiac/renal/neuro surveillance and complication management. **Genetic counseling** (autosomal dominant; mostly *de novo* with low recurrence, but 50% for an affected transmitting parent); prenatal/preimplantation and cascade testing possible once a familial variant is known.

14. Other species / natural disease. Mouse *Prkd1* (NCBI Gene 18760) is the conserved ortholog; no naturally occurring companion-animal disease reported. No zoonotic relevance.

15. Model organisms. Mouse is the principal model (F007): *Prkd1* knockout is embryonic lethal; tissue-specific deletion causes abnormal heart remodeling; a *Prkd1* transgenic model was used to study cardiogenesis/CHD (PMID: 38419169). Limitation: existing mouse data model loss-of-function, whereas the dominant human syndrome is gain-of-function — a **knock-in** of a patient missense allele would be required to fully recapitulate CHDED.

Limitations and Knowledge Gaps

- **Very small sample size.** Only ~8–10 molecularly confirmed dominant cases exist, so phenotype frequencies, penetrance, expressivity, and genotype–phenotype correlations are preliminary and subject to ascertainment bias.
- **Gain-of-function is inferred, not biochemically proven for patient variants.** The conclusion rests on gnomAD constraint (tolerance of LoF) and the clinical "gain-of-function syndrome" label. No published enzyme-kinetic or cellular assay directly demonstrates increased (or dominant-negative) kinase activity for p.Arg603His or p.Val712Met.
- **No patient-allele animal model.** Mouse data are from knockouts/transgenics modeling loss-of-function; they do not test the dominant missense mechanism.
- **The ITGB2/JAM2 calcification branch is a hypothesis.** It is proposed from gene-pathway overlap, not demonstrated for PRKD1 variants.
- **No epidemiology, QoL, survival, or natural-history data.** Prevalence/incidence are unknown; long-term outcomes are inferred from individual case follow-up.
- **No targeted therapy or clinical trial.** Management is entirely empirical/supportive.

Proposed Follow-up Experiments / Actions

1. **Functional validation of gain-of-function.** Express patient variants (p.Arg603His, p.Val712Met) in cells and measure in-vitro kinase activity, activation-loop autophosphorylation, and HDAC5 phosphorylation/nuclear export vs. wild-type and kinase-dead controls to directly test the gain-of-function vs. dominant-negative hypothesis.
2. **Knock-in animal model.** Generate a heterozygous *Prkd1* patient-allele knock-in mouse (or zebrafish) to test whether it recapitulates outflow-tract/septal defects and ectodermal/renal phenotypes — the decisive test of the dominant mechanism.
3. **Test the ITGB2/JAM2 branch.** Use co-immunoprecipitation, endothelial tight-junction/permeability assays, and blood–brain-barrier models with PRKD1 variants to confirm or refute the proposed calcification pathway.
4. **Patient registry and deep phenotyping.** Establish an international CHDED/TEBC registry (via GeneMatcher and syndromic-CHD consortia) to accumulate cases, quantify penetrance/expressivity, and define genotype–phenotype correlations.

5. **Developmental expression mapping.** Single-cell/spatial transcriptomics of PRKD1 in human and mouse embryonic heart (outflow tract, cardiac neural crest), skin, and kidney to pinpoint the vulnerable cell populations.
6. **Diagnostic standardization.** Ensure *PRKD1* is included in all syndromic-CHD and ectodermal-dysplasia gene panels; define a recommended multi-organ imaging/surveillance protocol (echocardiography, brain/renal imaging for calcification, EEG, vascular imaging for dissection risk).

Report generated from a 5-iteration autonomous investigation (15 confirmed findings, 36 papers reviewed). Evidence source types are indicated throughout: human clinical (case reports, exome cohort), model organism (mouse), in vitro (cardiomyocyte signaling), and computational (gnomAD constraint, GTEx expression).
